

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Case Study

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO3 – Precision (accurate feature extraction & analysis) (BL3)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	2024 AIML	Date:	11/8/2026

Code of Conduct

I M.MANASA(192425189)____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: m.manasa_____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Case Study: Feature Matching Under Transformations

Analysis of Traditional and Scale-Invariant Feature Matching

1. Case Description

Feature matching is a computer vision technique used to identify corresponding features between two images. In this case study, two images contain the same object, but the object appears at different scales and orientations. Traditional feature matching can become unreliable when the appearance of features changes because of scaling or rotation. Scale-invariant methods are designed to recognize distinctive features despite these transformations.

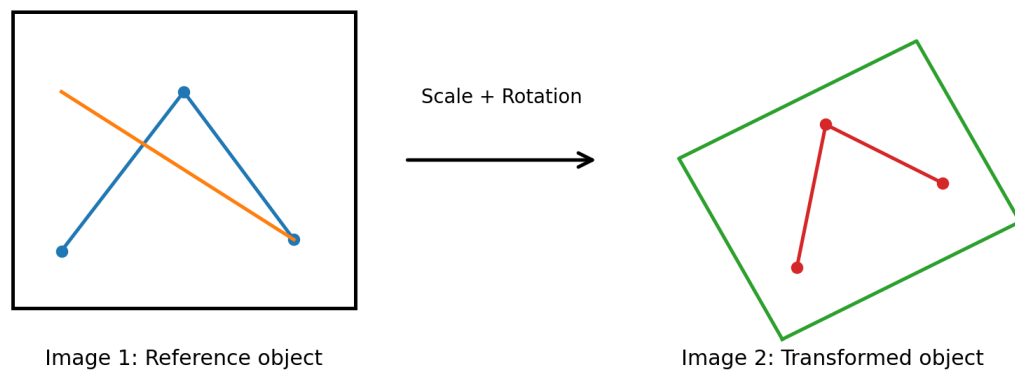


Figure 1. Illustrative example of scale and rotation transformations.

2. Problem Scenario

Consider a book, logo, building, or other object photographed from two different conditions. In the first image, the object may be large and upright. In the second image, the same object may appear smaller and rotated. A reliable matching system should still recognize that the two images contain corresponding parts of the same object.

3. Why Traditional Matching Can Fail

Traditional matching methods often depend strongly on the appearance of local image patterns. When the object is resized, the size of the local pattern changes. When the object is rotated, its orientation also changes. As a result, the same physical feature may no longer look sufficiently similar to the original feature, producing fewer correct matches or more incorrect matches.

4. Scale-Invariant Feature Matching

SIFT (Scale-Invariant Feature Transform) is a well-known feature detection and description method designed to provide robustness to scale and rotation changes. It detects distinctive keypoints, assigns an orientation to each keypoint, and constructs a descriptor that can be compared with descriptors from another image. OpenCV documents SIFT as a method for finding keypoints and descriptors, including orientation assignment and descriptor-based matching.

Illustrative Scale-Invariant Feature Matching Pipeline



Figure 2. Illustrative SIFT-style feature matching pipeline.

5. Comparison of Matching Methods

Parameter	Traditional Feature Matching	Scale-Invariant Matching
Scale change	Sensitive; matching can decrease	Designed to handle scale changes
Rotation	May be affected by rotation	Orientation assignment improves rotation robustness
Keypoints	May not remain consistent after scaling	Distinctive keypoints are detected across scales
Matching accuracy	Can decrease after transformation	Generally more reliable under scale/rotation changes
Robustness	Lower for transformed images	Higher for scale and rotation transformations
Applications	Best when images are very similar	Useful for object recognition and image matching

6. SIFT Working Steps

Keypoint Detection: Find distinctive points in the image at different scales.

Keypoint Localization: Refine the locations and remove weak or unsuitable points.

Orientation Assignment: Assign a dominant orientation to each keypoint so rotation changes can be handled.

Descriptor Generation: Represent the local neighborhood around each keypoint using a numerical descriptor.

Feature Matching: Compare descriptors from the two images and identify likely correspondences.

7. Example Analysis

Suppose a logo appears large and upright in Image 1 but small and rotated in Image 2. A traditional matcher may find only a limited number of correct correspondences because the feature appearance has changed. A scale-invariant method can use keypoints and orientation-aware descriptors to find corresponding regions more reliably.

8. Advantages of Scale-Invariant Matching

- Recognizes distinctive features at different sizes.
- Handles in-plane rotation more effectively.
- Provides useful local descriptors for image-to-image comparison.
- Improves robustness when the same object is captured under different viewpoints or camera distances.
- Can be used as a building block for object recognition, image registration, panorama creation, and image retrieval.

9. Applications

- Object recognition
- Image stitching and panorama creation
- Augmented reality
- Image retrieval
- Object tracking
- Image registration and alignment

10. Key Findings

The case study shows that changes in scale and orientation can reduce the reliability of traditional feature matching. Scale-invariant approaches are better suited to this scenario because their keypoint detection and descriptors are designed to remain useful under scale and rotation changes. However, performance can still depend on image quality, the distinctiveness of the object, and the amount of transformation.

11. Conclusion

Scale-invariant feature matching performs better than traditional matching when the same object is observed at different scales and orientations. By detecting distinctive keypoints, assigning orientations, and constructing robust local descriptors, methods such as SIFT can establish more reliable correspondences between transformed images.

12. References

1. OpenCV Documentation, "Introduction to SIFT (Scale-Invariant Feature Transform)."
https://docs.opencv.org/5.0/py_tutorials/py_features/py_sift_intro/py_sift_intro.html
2. OpenCV Documentation, "cv::SIFT Class Reference."
https://docs.opencv.org/4.5.4/d7/d60/classcv_1_1SIFT.html
3. Karami, E., Prasad, S., and Shehata, M., "Image Matching Using SIFT, SURF, BRIEF and ORB: Performance Comparison for Distorted Images," arXiv, 2017.

Note: Figures 1 and 2 are original illustrative diagrams created for this case study.